▸ FERNANDO BLANCÁ, NOEMÍ LUBOMIRSKY, AND RICARDO RODRIGUEZ, *Twist construction for BCI-algebras.*

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The objective of this talk is to present a unified approach that provides a deeper understanding of the classes of implicative algebras that can be represented by twist structures. We choose BCI-algebras as a starting point to do so because their logical counterpart constitutes a common implication fragment shared by many of the logics considered in the literature, such as classical logic, intuitionistic logic and the most important many-valued logics. They were first defined by Iséki in [1] precisely to describe fragments of the propositional calculus involving implication.

In line with [2], certain interior operators called *conuclei* will be central to obtaining the twist representations. Using them, we will provide two different representations for classes of algebras that generalize both involutive and non-involutive twist constructions over suitable BCI-algebras, together with a categorical interpretation of each construction. Finally, we will observe that our results concerning non-involutivity are connected with Rivieccio's work on the representation of quasi-Nelson implication algebras (QNI-algebras) via twist products of bounded nuclear Hilbert semigroups (nH-semigroups) [3].

[1] Iséki K, *An algebra related with a propositional calculus*, ***Proceedings of the Japan Academy*** (7-32 Ueno Park, Taito-ku, Tokyo 110-0007, Japan), (Shigekazu Nagata), vol. 42, Japan Academy, 1966, pp. 26–29.

[2] Busaniche M, Galatos N, Marcos M, *Twist structures and Nelson conuclei*, ***Studia Logica***, vol. 110 (2022), pp. 949–987.

[3] Rivieccio U, *Fragments of Quasi-Nelson: The Algebrizable Core*, ***Logic Journal of the IGPL***, vol. 30 (2022), no. 5, pp. 807–839.